SPRINT 3 - TEAMWORK PLAN

Sprint dates: 19 October – 1 November 2026

1. Sprint goal

The dataset is cleaned and frozen, the machine-learning rules and evaluation protocol are documented, and the team already has two simple baselines plus a first gradient-boosting model. This sprint will test two new models, compare all available results fairly, understand the main errors and prepare a clear handover to Sprint 4.

The goal is not to build the most complicated model. The goal is to find out which approach predicts energy consumption best and explain the results with evidence.

2. Rules everyone must follow

- Never modify the frozen source dataset; use separate derived files.
- Use the same evaluation periods, targets and forecast horizons for fair comparisons.
- Use validation to choose models. Do not use test results to tune or improve a model.
- Do not replace missing or unreliable energy values with zero.
- Keep the 2020–2023 period separate from the 2025 period.
- Use MAE as the main error metric and RMSE as a secondary metric.

3. Evaluation dates from Task 6

Check that the chosen dates have the expected data coverage before running experiments.

4. The five remaining tasks

TASK 1 — Build a Random Tree model and a Linear Regression model

Why? We already have a gradient-boosting model. Testing two different approaches will show whether they can make better predictions.

What to do:

- Use the agreed modelling dataset and, wherever possible, the same input variables as the existing model.
- Build a Random Tree model. If the intended algorithm is Random Forest rather than a single Decision Tree, clearly record which one is used.
- Build a Linear Regression model. Ridge Regression is an option only if the team agrees and records the choice.
- Train using the training period only. Use validation to choose settings; never tune with test results.
- Evaluate +1 hour, +6 hours and the next 24 hours where the implementation supports those horizons correctly.
- Save the code, model settings, predictions, MAE and RMSE.

Deliverable: Two runnable scripts/notebooks, saved predictions and a short results file for both models.

Finished when: Another team member can run both models and reproduce the recorded validation results.

Parallel work: Yes. One person builds Random Tree while another builds Linear Regression.

TASK 2 — Test one additional model (optional if time is limited)

Why? A third new model can add useful evidence, but it must not prevent the team from evaluating the main models properly.

What to do:

- Check which algorithm is already used by the existing gradient-boosting model.
- If feasible, test one additional model such as LightGBM only if it is not already the current model and adds a meaningful comparison.
- Prophet is optional; use it only if the team can explain why it fits the data and forecasting setup.
- Use the same training/validation periods, target, horizons and evaluation rules.
- Save predictions and calculate MAE/RMSE. If no extra model is tested, record why.

Deliverable: Comparable results for one extra model, or a short note explaining why it was not tested.

Finished when: The team has one additional set of results that can be compared fairly, or a documented decision to skip it.

Parallel work: Yes, after the shared dataset and evaluation rules are ready.

TASK 3 — Compare all models and investigate errors

Why? We need evidence to decide which model works best and understand when predictions are unreliable.

What to do:

- Collect results for B1 (same hour last week), B2 (weekday/hour median), B3 persistence if completed, the existing gradient-boosting model, Random Tree and Linear Regression.
- Include the optional model from Task 2 if available.
- Create one table with target, horizon, model, MAE, RMSE and evaluation period.
- Compare models on the same valid observations wherever possible.
- Choose the leading model using validation results. Once settings are fixed, run the final test evaluation; do not tune afterwards.
- Inspect the largest errors and note whether they occur at particular times, zones or data-quality problems.
- For 24-hour forecasts, report overall error and, if possible, error by lead time from hour 1 to hour 24.

Deliverable: A comparison table and a few clear graphs/examples showing the main prediction errors.

Finished when: The team can explain the best model by horizon, whether it beats the baselines and which limitations remain.

Parallel work: Partly. Prepare the table and graph templates early; complete them when model results are ready.

TASK 4 — Show how uncertain the predictions are

Why? A forecast is an estimate, not a guarantee. Users should understand how far actual consumption may be from the prediction.

What to do:

- Start with the model selected using validation results.
- Choose a simple, explainable method to estimate lower and upper prediction limits.
- Build the method without using test results to tune it.
- Measure the percentage of actual values that fall inside the prediction band.
- Create a graph showing actual consumption, the prediction and the lower/upper limits.
- Explain the chosen coverage level and the observed coverage. If it cannot be completed reliably, document the method and remaining work.

Deliverable: A prediction-band graph and coverage summary, or a short note explaining what remains to be done.

Finished when: The team can explain how the band was created and how often it contained actual values, or clearly document why it is unfinished.

Parallel work: Research can start earlier; implementation follows selection of a sufficiently evaluated model.

TASK 5 — Prepare the final report and Sprint 4 handover

Why? The results must be understandable to the sponsor and easy for another team member to reproduce.

What to do:

- Write a short report in English explaining the objective, data periods and evaluation rules.

- List only the models actually tested: existing gradient boosting, Random Tree, Linear Regression and any extra model completed.
- Include the comparison table, MAE/RMSE results and main error graphs.
- Explain which model is best for each tested horizon and whether it beats the simple baselines.
- Describe data limitations and prediction-band results if available.
- Organise code, configuration, predictions, metrics and experiment notes so another team member can reproduce the work.
- Recommend next steps for Sprint 4, potentially investigating unusual energy consumption and possible energy waste.

Deliverable: Final English report, organised code/results, experiment notes and a clear Sprint 4 recommendation.

Finished when: The team can present its conclusions and another member can reproduce the main evaluation.

Parallel work: Prepare the report template and experiment log throughout the sprint; write conclusions after Task 3.

5. Sprint 3 — Definition of Done

- Random Tree and Linear Regression are trained and evaluated using the agreed protocol.
- Existing gradient boosting and simple baselines are included where results are available.
- An additional model is tested if feasible, or the decision to skip it is documented.
- Models are compared using MAE/RMSE on comparable observations.
- Validation is used for model selection; test is used only for final evaluation.
- Main errors and data limitations are explained.
- Prediction uncertainty is evaluated or remaining work is documented.
- Report, code and results are organised for reproduction and Sprint 4 handover.

Final message to the team

Keep the work simple and traceable. Every task must produce a file, table, graph or written decision. Do not claim a model is better unless the comparison supports it. Reliable results are more valuable than testing many algorithms without enough time to evaluate them.